

This is a copy of a conversation between ChatGPT & Anonymous.

Report conversation

I'm writing some self-study guides for students on my technical course. For context, the course involves working with python notebooks and python modules. Students might work in Python modules, or in ipynb notebooks. Students will typically work in Cursor. For package management, we're using uv. Students may be on Windows PC or Mac.

Please write a thorough, comprehensive tutorial that gives the very basics of python for someone completely new to python. They will have a Notebook in Cursor that they can use to experiment with your instructions. Note that they can't use "input" as it doesn't work in a cursor notebook.

Part 1: Introduction - what is python, the python interpreter

Part 2: Strings and numbers and booleans. The print statement

Part 3: Variables. Setting a variable. printing a variable.

Part 4: A simple if statement with some examples.

Part 5: A simple for loop with a range, with examples.

Part 6. Exercises

Part 7. More things to try: while loops. lists. dicts.

Please make it very comprehensive and for a complete beginner.

Great! Here's a comprehensive beginner-friendly Python tutorial written specifically for students using Python notebooks in Cursor with `uv` for environment management. This guide assumes no prior programming experience and avoids using `input()` since it doesn't work in Cursor notebooks.



Python for Total Beginners – Part 1 to 7

This guide is your first step into the world of programming using **Python**! You'll be writing code directly in a Python notebook using **Cursor** (a VSCode-based editor). You can run each cell by pressing `Shift + Enter`. Feel free to tweak examples and try your own experiments!

✓ Part 1: What Is Python?

Python is one of the most popular programming languages in the world. It's known for being:

- Easy to read
- Easy to write
- Extremely powerful

Python is used for web development, data science, automation, AI, games, and more.

🔍 What Is a Python Interpreter?

The **Python interpreter** is the program that reads and runs your Python code. In a notebook, every cell is sent to the interpreter when you run it. The interpreter understands your code line by line and gives back results.

abc Part 2: Strings, Numbers, Booleans & the `print()` Function

Python works with different **data types**. Let's look at the basics:

📌 Strings

A **string** is text in quotes.

```
python
```

```
print("Hello, world!")  
print('Python is fun!')
```

📄 Copy code

1 2 3 4 Numbers

Python supports both **integers** (whole numbers) and **floats** (decimals):

```
python
```

```
print(42)           # an integer  
print(3.14)         # a float
```

```
print(2 + 3)    # math!
```

 Copy code

✓ Booleans

Booleans are either `True` or `False` :

python

```
print(True)
print(False)
print(5 > 3)    # True
print(2 == 4)   # False
```

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Part 3: Variables

You can **store** values in **variables** and use them later.

python

```
greeting = "Hello there!"
age = 25
is_student = True
```

 Copy code

Printing variables

python

```
print(greeting)
print(age)
print(is_student)
```

 Copy code

You can even combine text and variables:

python

```
name = "Taylor"
print("Hello, " + name + "!")
```

 Copy code

Part 4: If Statements

An **if statement** lets your program make decisions.

```
python
```

```
age = 18

if age >= 18:
    print("You're an adult!")
```

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You can also add `else` and `elif` :

```
python
```

```
age = 16

if age >= 18:
    print("You're an adult!")
elif age >= 13:
    print("You're a teenager!")
else:
    print("You're a child!")
```

 Copy code

Use **indentation (spaces)** to show which lines belong to the `if` .

Part 5: For Loops with `range()`

Loops let you repeat code.

```
python
```

```
for i in range(5):
    print("i is", i)
```

 Copy code

Try changing the range:

```
python
```

```
for number in range(1, 6):
    print(number * number)
```

 Copy code

Explanation

- `range(5)` goes from 0 to 4 (five numbers)
- `range(1, 6)` goes from 1 to 5



Part 6: Exercises

Try these in your notebook:

1. Print your name 3 times using a loop.
 2. Store your age in a variable and print a different message depending on whether you're under 18 or not.
 3. Make a loop that prints the squares of the numbers 1–10.
 4. Store a `favorite_color` variable and print a custom message depending on what color it is.
 5. Try combining strings with variables using `+`.
-



Part 7: More Things to Try



While Loops

Repeat something **while** a condition is true.

```
python
```

```
x = 0
```

```
while x < 5:  
    print("x is", x)  
    x = x + 1
```

Copy code



Lists

Lists store **multiple values**:

```
python
```

```
colors = ["red", "green", "blue"]
```

≡ ChatGPT 5.2 Thinking ✓

you can loop through them.

```
python
```

```
for color in colors:  
    print("I like", color)
```

Copy code



Dictionaries

Dictionaries store **key-value pairs**:

python

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```
person = {  
    "name": "Taylor",  
    "age": 30,  
    "is_student": False  
}  
  
print(person["name"])
```

You can loop through them too:

python

 Copy code

```
for key in person:  
    print(key, "=", person[key])
```

That's it for now! 🎉 You've just learned the core of Python.

Up next, we'll dive deeper into **functions**, **classes**, **modules**, and more. But for now, **practice these basics** until they feel easy.

Let me know if you'd like a Part 8 for next steps, or want this exported into a notebook format (`.ipynb`) or Markdown!